

Assignment - 3

Date:

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1. Determine multiplicative modulo for each of following integers: 4, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, 26

a) 3^{-1}

$$\begin{aligned} 26 &= 3(8) + 2 \\ 3 &= 2(1) + 1 \\ 1 &= 3(2) - 3 + 2(-1) \\ 1 &= 3 + 2(-1) + 3(-1) \\ 3(1) + 26(-1) &= 1 \\ 3^{-1} &= 1 \end{aligned}$$

b) 7^{-1}

$$\begin{aligned} 26 &= 7(3) + 5 \\ 7 &= 5(1) + 2 \\ 5 &= 2(2) + 1 \\ 26 + 7(3) + 5(-1) &= 1 \\ 26 + 7(3) + 7(-2) + 5(2) &= 1 \\ 26 + 7(-5) + 5(2) &= 1 \\ 26 + 7(-5) + (26 + 7(-3))(2) &= 1 \\ 26 + 7(-5) + 26(2) + 7(-6) &= 1 \\ 26(3) + 7(-11) &= 1 \end{aligned}$$

b) 5^{-1}

$$\begin{aligned} 26 &= 5(5) + 1 \\ 26 + 5(-5) &= 1 \\ 26^{-5} &= 21 \\ 5^{-1} &= 21 \end{aligned}$$

a) 9^{-1}

$$\begin{aligned} 26 &= 9(2) + 8 \\ 9 &= 8(1) + 1 \\ 8 &= 26 + 9(-2) \\ 9 + 8(-1) &= 1 \end{aligned}$$

$$\begin{aligned} 9 + [26 + 9(-2)](-1) &= 1 \\ 9 + [26(-1) + 9(2)] &= 1 \\ 9(3) + 26(-1) &= 1 \\ 9^{-1} &= 3 \end{aligned}$$

e) 11^{-1}

$$\begin{aligned} 26 &= 11(2) + 4 \\ 11 &= 4(2) + 3 \\ 4 &= 3(1) + 1 \\ 26 + 11(-2) &= 4 \\ 11 + 4(-2) &= 3 \\ 4 + 3(-1) &= 1 \end{aligned}$$

$$11^{-1} = 19$$

f) 15^{-1}

$$\begin{aligned} 26 &= 15(1) + 11 \\ 15 &= 11(1) + 4 \\ 11 &= 4(2) + 3 \\ 4 &= 3(1) + 1 \\ 15 + 11(-1) + [11 + 4(-2)](-1) &= 1 \\ 15 + 11(-1) + 11(-1) + 4(2) &= 1 \\ 15 + 11(-2) + 4(2) &= 1 \\ 15 + [26 + 15(-1)](-2) + [15 + 11(-1)](2) &= 1 \\ 15 + 26(-2) + 15(2) + 11(-2) &= 1 \\ 15(5) + 26(-2) + [26 + 15(-1)](-2) &= 1 \\ 15(5) + 26(-2) + 26(-2) + 15(2) &= 1 \\ 15(1) + 26(-4) &= 1 \\ 15^{-1} &= 1 \end{aligned}$$

g) 17^{-1}

$$\begin{aligned} 26 &= 17(1) + 9 & 26 + 17(-1) &= 9 \\ 17 &= 9(1) + 8 & 17 + 9(-1) &= 8 \\ 9 &= 8(1) + 1 & 9 + 8(-1) &= 1 \\ 26 + 17(-1) + [17 + 9(-1)](-1) &= 1 \\ 26 + 17(-1) + 17(-1) + 9(1) &= 1 \\ 26 + 17(-1) + 17(-1) + 26 + 17(-1) &= 1 \\ 26(2) + 17(-3) &= 1 \\ 26 - 3 &= 23 \\ \underline{\underline{17^{-1} = 23}} \end{aligned}$$

h) 19^{-1}

$$\begin{aligned} 26 &= 19(1) + 7 & 26 + 19(-1) &= 7 \\ 19 &= 7(2) + 5 & 19 + 7(-2) &= 5 \\ 7 &= 5(1) + 2 & 7 + 5(-1) &= 2 \\ 5 &= 2(2) + 1 & 5 + 2(-2) &= 1 \\ 19 + 7(-2) + [7 + 5(-1)](-2) &= 1 \\ 19 + [26 + 19(-1)](2) + 19(2) + [19 + 7(-2)](2) &= 1 \\ 19(5) + 26(-4) + 19(2) + 7(-4) &= 1 \\ 19(7) + 26(-4) + [26 + 19(-1)](-4) &= 1 \\ 19(7) + 26(-4) + 26(-4) + 19(4) &= 1 \\ 19(11) + 26(-8) &= 1 \\ \underline{\underline{19^{-1} = 11}} \end{aligned}$$

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i) 21^{-1}

$$\begin{aligned} 26 &= 21(1) + 5 & 26 + 21(-1) &= 5 \\ 21 &= 5(4) + 1 & 21 + 5(-4) &= 1 \\ 21 + [26 + 21(-1)](-4) &= 1 \\ 21 + 26(-4) + 21(4) &= 1 \\ 21(5) + 26(-4) &= 1 \\ \underline{\underline{21^{-1} = 5}} \end{aligned}$$

k) 25^{-1}

$$\begin{aligned} 26 &= 25(1) + 1 \\ 26 + 25(-1) &= 1 \\ 26 - 1 &= 25 \\ \underline{\underline{25^{-1} = 25}} \end{aligned}$$

j) 23^{-1}

$$\begin{aligned} 26 &= 23(1) + 3 & 26 + 23(-1) &= 3 \\ 23 &= 3(7) + 2 & 23 + 3(-7) &= 2 \\ 3 &= 2(1) + 1 \\ 26 + 23(-1) + [23 + 3(-7)](-1) &= 1 \\ 26 + 23(-1) + 23(-1) + 3(7) &= 1 \\ 26 + 23(-2) + 3(7) &= 1 \\ 26 + 23(-2) + [26 + 23(-1)](7) &= 1 \\ 26 + 23(-2) + 26(7) + 23(-7) &= 1 \\ 26(8) + 23(-9) &= 1 \\ 26 - 9 &= 17 \\ \underline{\underline{23^{-1} = 17}} \end{aligned}$$

2. Choose any string of your choice to encrypt using the affine cipher with key values: multiplicative 5 & Addictive 8

$$\rightarrow CT = (P * K1 + K2) \bmod 26$$

$$K1 = 5, K2 = 8$$

PT = ABHI

$$A = (0 * 5 + 8) \bmod 26 = 8 \bmod 26 = 8 = I$$

$$B = (1 * 5 + 8) \bmod 26 = 13 \bmod 26 = 13 = N$$

$$H = (7 * 5 + 8) \bmod 26 = 43 \bmod 26 = 17 = R$$

$$I = (8 * 5 + 8) \bmod 26 = 48 \bmod 26 = 22 = W$$

CT = INRW

3. Encrypt "HI" using affine cipher with $a=11$ & $b=5$

$$\rightarrow C = (P * a + b) \bmod 26, a=11; b=5$$

$$H = (7 * 11 + 5) \bmod 26 = 82 \bmod 26 = 4 = E$$

$$I = (8 * 11 + 5) \bmod 26 = 93 \bmod 26 = 15 = P$$

CT = EP

4. The cipher text 'UCR' was encrypted using affine function $9x+2 \bmod 26$. Find plain text

$$\rightarrow CT = (9PT + 2) \bmod 26$$

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$$PT = a^{-1}(C - b) \bmod 26 \quad a=9, b=2$$

By inverse, $9^{-1} = 3$

$$U = 3(20 - 2) \bmod 26 = 54 \bmod 26 = 2 = C$$

$$C = 3(2 - 2) \bmod 26 = 0 \bmod 26 = 0 = A$$

$$R = 3(17 - 2) \bmod 26 = 45 \bmod 26 = 19 = T$$

PT = CAT

5. Decrypt the cipher text using Affine cipher:

KDYXJUWDFXZY using the key 3 (Multiplication) & 4.

$$\rightarrow PT = a^{-1}(C - b) \bmod 26, a=3, b=4$$

$3^{-1} = 9$ by $26 = 3(8) + 2$

$$3 = 2(1) + 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1 \Rightarrow \underline{3^{-1} = 9}$$

$$K = 9(10 - 4) \bmod 26 = 54 \bmod 26 = 2 = C$$

$$D = 9(3 - 4) \bmod 26 = -9 \bmod 26 = 26 - 9 = 17 = R$$

$$Y = 9(24 - 4) \bmod 26 = 180 \bmod 26 = 24 = Y$$

$$X = 9(23 - 4) \bmod 26 = 171 \bmod 26 = 15 = P$$

$$J = 9(9 - 4) \bmod 26 = 45 \bmod 26 = 19 = T$$

$$V = 9(20 - 4) \bmod 26 = 144 \bmod 26 = 14 = D$$

$$\begin{aligned}
 W &= 9(22-4) \bmod 26 = 162 \bmod 26 = 6 = G \\
 D &= 9(8-4) \bmod 26 = -9 \bmod 26 = 26-9 = 17 = O \\
 E &= 9(4-4) \bmod 26 = 0 \bmod 26 = 0 = G \\
 X &= 9(23-4) \bmod 26 = 171 \bmod 26 = 15 = P \\
 Z &= 9(25-4) \bmod 26 = 189 \bmod 26 = 7 = H \\
 Y &= 9(24-4) \bmod 26 = 18 \bmod 26 = 24 = Y
 \end{aligned}$$

PT = CRYPTOGRAPHY

6) Use $K_1 = 13$ (Multiplier), $K_2 = 4$ (Even though it is not a valid key), Encrypt the string "input" & "alter" using key & find what is the issue.

$$C = (P \times K_1 + K_2) \bmod 26 \quad K_1 = 13, K_2 = 4$$

13 is not a coprime of 26. So we won't be able to find inverse of 13 mod 26. That's why it is not a valid key for Affine Cipher.

$$\begin{aligned}
 I &= (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 = E \\
 N &= (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 = R \\
 P &= (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 = R \\
 U &= (20 \times 13 + 4) \bmod 26 = 264 \bmod 26 = 4 = E \\
 T &= (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R \\
 A &= (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 = E \\
 L &= (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 = R \\
 T &= (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R \\
 E &= (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 = E \\
 T &= (17 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 = R
 \end{aligned}$$

PT: ERROR

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Both the different plain text produce the same cipher text. The encryption is not 1-to-1 & decryption can't be possible coz key is 13 which is not coprime of 26.

7) Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$.

$$\rightarrow K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$$

$$\text{Det of } K = |3 \times 7 - 2 \times 5| = |21 - 10| = 11$$

$$11^{-1} \bmod 26 = 11(2) + 4 \quad 26 + 11(-2) = 4$$

$$11 = 4(2) + 3 \quad 11 + 4(-2) = 3$$

$$4 = 3(1) + 1 \quad 4 + 3(-1) = 1$$

$$[26 + 11(-2)] + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\underline{\underline{11^{-1} = 19}}$$

Adj matrix of $K = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find Inverse of the following matrix whose entries are considered mod 26. $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$\rightarrow K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \quad |K| = |11 \times 3 - 13 \times 2| = |33 - 26| = 7$$

$$\begin{aligned} 7^{-1} \Rightarrow 26 &= 7(3) + 5 & 5 &= 26 + 7(-3) \\ 7 &= 5(1) + 2 & 2 &= 7 + 5(-1) \\ 5 &= 2(2) + 1 & 1 &= 5 + 2(-2) \end{aligned}$$

$$\begin{aligned} 5 + [7 + 5(-1)](-2) &= 1 \\ 26 + 7(-3) + 7(-2) + 5(2) &= 1 \\ 26 + 7(-5) + [26 + 7(-3)](2) &= 1 \\ 26 + 7(-5) + 26(2) + 7(-6) &= 1 \\ 26(3) + 7(-11) &= 1 \end{aligned}$$

$$26 - 11 = 15$$

$$7^{-1} = 15$$

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$$\text{Adj of } K = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 45 & -195 \\ -3 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9) Is the key valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\rightarrow |K| = |2 \times 21 - 14 \times 13| = |42 - 182| = -140$$

$$140 \pmod{26} = 16$$

$$K = 16$$

16 is not a coprime of 26. So key is not valid

10) Encrypt using 2x2 Hill cipher for PT: 'SOLVED' using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$\rightarrow K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \quad |K| = |3 \times 11 - 8 \times 7| = |33 - 56| = -23 \pmod{26} = 3$$

(coprime of 26)

$$\text{SOLVED} \Rightarrow C = (K \times P) \pmod{26}$$

$$S = \begin{bmatrix} 18 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 54 + 112 \\ 126 + 44 \end{bmatrix} \pmod{26} = \begin{bmatrix} 166 \\ 170 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \rightarrow K \\ 20 \rightarrow V \end{bmatrix}$$

$$L = \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} \pmod{26} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 \rightarrow T \\ 22 \rightarrow W \end{bmatrix}$$

$$E = \begin{bmatrix} 4 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \pmod{26} = \begin{bmatrix} 12 + 24 \\ 21 + 33 \end{bmatrix} \pmod{26} = \begin{bmatrix} 36 \\ 54 \end{bmatrix} \pmod{26} = \begin{bmatrix} 10 \rightarrow K \\ 28 \rightarrow J \end{bmatrix}$$

CT = KOTWKEJ